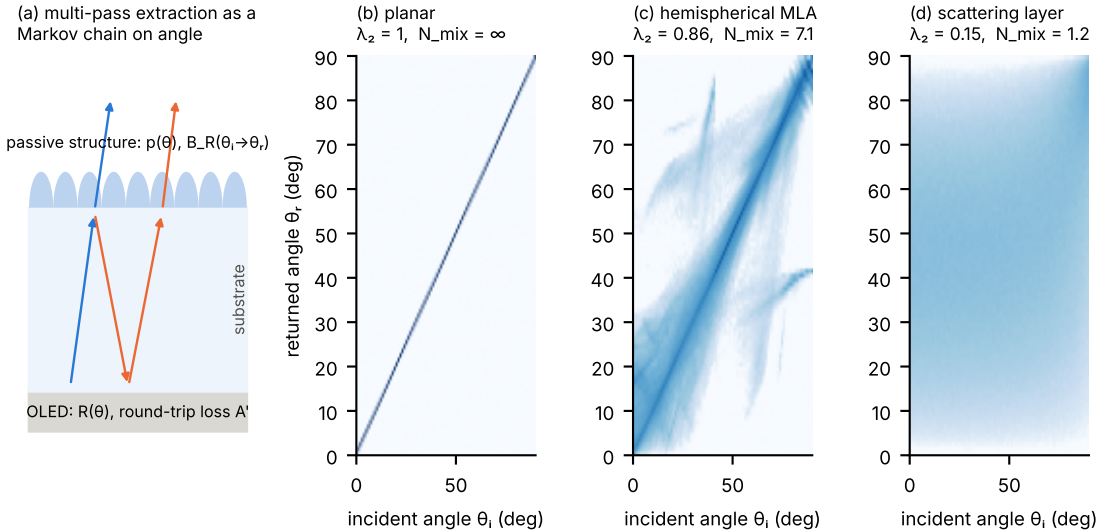

(a) multi-pass extraction as a Markov chain on angle
passive structure: p(θ), B_R(θi→θr)
substrate
OLED: R(θ), round-trip loss A'
(b) planar
λ2 = 1, N_mix = ∞
(c) hemispherical MLA
λ2 = 0.86, N_mix = 7.1
(d) scattering layer
λ2 = 0.15, N_mix = 1.2
returned angle θr (deg)
incident angle θi (deg)
0
10
20
30
40
50
60
70
80
90